

# Recovering the familiar angular momentum operators

The two-angle limit of the body-fixed framework

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# Why a supplement?

In the main deck the body-fixed operators

$$\hat{J}_\alpha = -i\hbar \sum_s (E^{-1})_{s\alpha} \partial_{\Omega_s} \quad (\phi, \theta, \chi) \quad (4.24)$$

look unlike the  $\hat{L}$  of introductory QM — three Euler angles, body-fixed (anomalous sign), and the zyz-fixed trig coefficients.

## Claim of this supplement

Choose the *familiar* setting — a single orientation described by **two** angles  $(\theta, \phi)$ , i.e. the zyz convention with the third angle  $\chi$  dropped — and the whole machinery collapses to the textbook operators

$$\hat{L}_z = -i\hbar \partial_\phi, \quad \hat{L}_x, \quad \hat{L}_y, \quad \hat{L}^2.$$

We derive them in full below.

## The two-angle setting: a point on a sphere / linear rotor

A point mass (or a linear/axially symmetric body) has *no* orientation about its own axis, so the third Euler angle  $\chi$  is redundant: two angles  $(\theta, \phi)$  suffice. In the  $zyz$  convention with  $\chi = 0$ ,

$$\mathbf{S}(\phi, \theta) = R_z(\phi) R_y(\theta), \quad \hat{\mathbf{n}}(\theta, \phi) = \mathbf{S}(\phi, \theta) \hat{\mathbf{z}} = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix},$$

— exactly the spherical-polar direction. The configuration space is the unit sphere  $S^2$ , with the rotationally invariant measure  $\sin \theta \, d\theta \, d\phi$  (the two-angle version of the deck's  $\sin \theta \, d\phi \, d\theta \, d\chi$ ).

*Goal: get the angular momentum operators on functions  $\psi(\theta, \phi)$ , in full.*

## Angular momentum as $\mathbf{r} \times \hat{\mathbf{p}}$ in spherical coordinates

Start from the definition  $\hat{\mathbf{L}} = \mathbf{r} \times \hat{\mathbf{p}} = -i\hbar \mathbf{r} \times \nabla$ , and write the gradient in the spherical orthonormal frame:

$$\nabla = \hat{\mathbf{r}} \partial_r + \hat{\boldsymbol{\theta}} \frac{1}{r} \partial_\theta + \hat{\boldsymbol{\phi}} \frac{1}{r \sin \theta} \partial_\phi, \quad \mathbf{r} = r \hat{\mathbf{r}}.$$

Then, since  $\hat{\mathbf{r}} \times \hat{\mathbf{r}} = 0$ ,

$$\mathbf{r} \times \nabla = r \hat{\mathbf{r}} \times \left( \hat{\boldsymbol{\theta}} \frac{1}{r} \partial_\theta + \hat{\boldsymbol{\phi}} \frac{1}{r \sin \theta} \partial_\phi \right) = (\hat{\mathbf{r}} \times \hat{\boldsymbol{\theta}}) \partial_\theta + (\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}}) \frac{1}{\sin \theta} \partial_\phi.$$

With  $\hat{\mathbf{r}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}$  and  $\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}} = -\hat{\boldsymbol{\theta}}$ ,

$$\hat{\mathbf{L}} = -i\hbar \left( \hat{\boldsymbol{\phi}} \partial_\theta - \hat{\boldsymbol{\theta}} \frac{1}{\sin \theta} \partial_\phi \right).$$

No radial derivative survives — angular momentum is purely angular.

## Project onto the space-fixed axes

The spherical unit vectors in Cartesian components are

$$\hat{\theta} = (\cos \theta \cos \phi, \cos \theta \sin \phi, -\sin \theta), \quad \hat{\phi} = (-\sin \phi, \cos \phi, 0).$$

Dotting  $\hat{\mathbf{L}} = -i\hbar(\hat{\phi} \partial_\theta - \hat{\theta} \frac{1}{\sin \theta} \partial_\phi)$  with  $\hat{\mathbf{z}}$  (then  $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ ):

$$\hat{L}_z = -i\hbar \left( \underbrace{(\hat{\phi})_z}_0 \partial_\theta - \underbrace{(\hat{\theta})_z}_{-\sin \theta} \frac{1}{\sin \theta} \partial_\phi \right) = \boxed{-i\hbar \partial_\phi}$$

$$\hat{L}_x = -i\hbar \left[ (-\sin \phi) \partial_\theta - (\cos \theta \cos \phi) \frac{1}{\sin \theta} \partial_\phi \right] = i\hbar (\sin \phi \partial_\theta + \cot \theta \cos \phi \partial_\phi),$$

$$\hat{L}_y = -i\hbar \left[ (\cos \phi) \partial_\theta - (\cos \theta \sin \phi) \frac{1}{\sin \theta} \partial_\phi \right] = i\hbar (-\cos \phi \partial_\theta + \cot \theta \sin \phi \partial_\phi).$$

These are exactly the orbital angular momentum operators of introductory QM.

## Ladder operators and the Casimir $\hat{L}^2$

Combining the two, the  $e^{\pm i\phi}$  structure emerges (using  $\cos \phi \pm i \sin \phi = e^{\pm i\phi}$ ):

$$\hat{L}_{\pm} \equiv \hat{L}_x \pm i \hat{L}_y = \hbar e^{\pm i\phi} (\pm \partial_{\theta} + i \cot \theta \partial_{\phi}).$$

The Casimir  $\hat{L}^2 = \hat{L}_z^2 + \frac{1}{2}(\hat{L}_+ \hat{L}_- + \hat{L}_- \hat{L}_+)$  collapses to

$$\hat{L}^2 = -\hbar^2 \left[ \frac{1}{\sin \theta} \partial_{\theta} (\sin \theta \partial_{\theta}) + \frac{1}{\sin^2 \theta} \partial_{\phi}^2 \right]$$

— precisely the **Laplace–Beltrami operator on the unit sphere** (the angular part of  $\nabla^2$ ), Hermitian on  $\sin \theta d\theta d\phi$ . Its eigenfunctions are the spherical harmonics,

$$\hat{L}^2 Y_l^m = \hbar^2 l(l+1) Y_l^m, \quad \hat{L}_z Y_l^m = \hbar m Y_l^m.$$

## Connection back to the main deck

Everything here is the  $\chi$ -free special case of the general framework:

- **Same convention.**  $\mathbf{S} = R_z(\phi)R_y(\theta)R_z(\chi)$  with  $\chi$  dropped is  $R_z(\phi)R_y(\theta)$  — the standard spherical-polar parametrization. Choosing it is what makes the operators come out familiar.
- **Same clean rotational block.**  $\hat{L}^2$  is the rotational Laplace–Beltrami operator on  $\sin \theta \, d\theta \, d\phi$ , Hermitian with *no* pseudopotential — the two-angle instance of the deck’s rigid-rotor result  $\frac{1}{2}\mu_{\alpha\beta}\hat{J}_\alpha\hat{J}_\beta$ .
- **Same  $1/\sin \theta$ .** The “unusual” coefficients already appear here in  $\hat{L}_x, \hat{L}_y$ ; they are the coordinate singularity at the poles, not an artifact.
- **Normal vs anomalous.** These  $\hat{L}_\alpha$  are *space-fixed* (a point particle has no body frame), so  $[\hat{L}_\alpha, \hat{L}_\beta] = +i\hbar\epsilon_{\alpha\beta\gamma}\hat{L}_\gamma$  — the ordinary sign. The anomalous sign of the deck’s  $\hat{J}_\alpha$  appears only for genuinely *body-fixed* components (three angles).

*The familiar operators are the special case; the general  $\hat{J}_\alpha$  (4.24) are their asymmetric-top generalization.*